What a buyer can learn from product, label and price data, and where to be careful

Same product, different colourway or department: found automatically.

Wrong labels: caught for review, but not safe to auto-correct.

Price architecture and markdown timing: reconstructed from sales alone.

Everything here is a simulated catalogue with a known answer key. It shows the method works, not what any retailer's market looks like.

Matching: finding the same style across colourways and departments

Text-only matching puts the true match first for 60% of products and finds 84% of all true matches in the top 10.

Adding a simulated image signal moved ranking quality from 0.700 to 0.710: too small to claim a gain from one run.

Real image embeddings (CLIP) have not been tested yet.

Method	NDCG@10	Match first
Text only	0.700	60%
Text + image (simulated)	0.710	60%

Simulated data (stylematch twin). Not real retailer data.

Labels: colour and type are reliable, department is not

Department is the weak spot: many descriptions never say who the item is for, so men's and women's items get swapped.

Flagging disagreeing labels catches 95% of planted department errors but only 19% of flags are real errors. Use it as a review list.

Product type is easy in this simulation; do not read 99.8% as a real-world benchmark.

Attribute	Accuracy
Product Type	99.8%
Colour	91.6%
Department	75.7%

Simulated data (stylematch twin). Not real retailer data.

Price architecture: where each category sits and how often it is marked down

First prices are reconstructed from sales, not read from a price file.

In this simulation categories look almost the same by construction (similar prices and markdown rates), so there is no real category story here. The view is ready for real data.

Category	Median price	Marked down	Median depth
Coat	25	52%	40%
Dress	24	49%	30%
Shoes	24	46%	39%
Skirt	25	50%	31%
Sweater	25	50%	31%
Swimwear	24	47%	35%
T-shirt	26	53%	35%
Trousers	23	49%	35%

Simulated data (stylematch twin). Not real retailer data.

Markdowns: detectable from sales, and slow starters go first

Markdown detection: no false alarms and 89% of planted markdowns found; timing off by 0.4 weeks on average.

Sales in the first four weeks predict when a markdown arrives: error 3.6 weeks against 5.7 using the category average.

Caveat: this relationship was planted in the simulation. On real data it must be re-tested.

What this data cannot support

Forecasts: the simple recent-average beat seasonal models (error 12% vs 46%) because products launch and fade within the window. Two years is too short to learn seasons.

No competitor prices, stock levels or web-crawled listings, so sell-through is a proxy and not sell-out.

No causal price-elasticity claims: the simulation's price response is an assumption.

Next step: run on real catalogue data with real image embeddings and compare against these baselines.